

Lorraine (Thi An Khanh) Nguyen

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EDUCATION

Bryn Mawr College

Bachelor of Arts in Computer Science, Minor in Statistics

Bryn Mawr, PA

Expected May 2028

Relevant Coursework: Intro to Data Structures, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Foundations of Data Science, Research Methods & Statistics

EXPERIENCE

Student Summer Researcher

AlgoArt

May 2026 – Aug 2026

Bryn Mawr, PA

- Co-authored the algTartan drawing tool in JavaScript and HTML5 Canvas with Professor Michael Wehar, engineering algorithmic layout controls including dynamic spacing gaps, axis isolation toggles, and randomized Fisher-Yates thread sorting.
- Engineered a modular Options panel in the Creator Studio and optimized color processing from mathematical filters to pre-generated hex code dictionaries, achieving $O(1)$ constant-time lookups that eliminated real-time rendering delays and prevented browser memory crashes during large-scale algorithmic weaves.

Help Desk Student Technician

Bryn Mawr College

Sep. 2025 – Present

Bryn Mawr, PA

- Resolve 40–50 hardware, software, and network issues weekly in collaboration with shift-mates and supervisors.
- Diagnose technical problems through documentation analysis, structured debugging, and clear communication with non-technical users.

Discrete Mathematics Teaching Assistant

Haverford College

Sep. 2025 – Dec. 2025

Haverford, PA

- Supported students in combinatorics, logic, and induction, reinforcing structured reasoning used in computer science and quantitative problem-solving.
- Led weekly sessions and graded proofs for correctness and rigor, requiring clear communication and attention to detail.

PROJECTS

Obsidian Plugin | *TypeScript, Node.js, Obsidian API, Git, Markdown, esbuild*

July 2025 – Present

- Develop a modular plugin that automatically links related notes using text parsing, event-driven logic, and maintainable component design across a vault of 500+ markdown notes.
- Improve caching workflow performance by approximately 30% by streamlining data access, eliminating noticeable lag and reducing redundant operations during large-scale text parsing.

Resume Keyword Analyzer | *Python, Streamlit, PyPDF, Git, spaCy, rapidfuzz*

March 2025 – Present

- Develop a tool that parses resumes and job descriptions across PDF, DOCX, and TXT formats and analyzes keyword alignment using NLP-based extraction and fuzzy concept matching.
- Build a modular processing pipeline for structured data processing and database-like document parsing for structured analysis, interactive review, and clear presentation of results.

Data Science Analysis (Coursework) | *Python, Pandas, NumPy, Matplotlib*

Jan 2026 - Present

- Analyzed datasets using exploratory data analysis, statistical modeling, and data visualization in Python.
- Applied machine learning concepts including Bayesian modeling and MCMC-based inference to interpret uncertainty

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, R, HTML/CSS

Tools/Frameworks: Git, Node.js, Docker, Streamlit, Obsidian API, HTML5 Canvas, JUnit, VS Code

Libraries: Pandas, NumPy, spaCy, NLTK, PyPDF, rapidfuzz

Concepts: Modular development, object-oriented design, algorithmic design, memory optimization, state management, debugging